



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251

Spring Semester 2025

Instructor: Usman Salahuddin

Hours: Thurs

(3:50 PM – 5:05 PM)

Class Location E-011

Office hour: Thu 9:45 – 11:45 AM

Office: C-103

Email: usman.salahuddin@sse.habib.edu.pk

USMAN SALAHUDDIN

WHAT DO I DO?

Renewable energy | Electric vehicles | Catalysis | Nanomaterials | Plastic recycling | Brain machine interfaces

WHAT HAVE I STUDIED?

PhD (Fulbright Scholar) | Material Science | University of Connecticut, USA | 2018 - 23

Masters | Energy Systems Engineering | NUST, Pakistan | 2015 - 17

Bachelors | Chemical Engineering | NUST, Pakistan | 2010 - 14

WHERE ELSE HAVE I WORKED?

Advanced Manufacturing | Additive manufacturing | Hartford, USA | 2023

R/GA Ventures | Technical support for startups | Storrs, USA | 2023

Tesla Motors | R&D Cell Engineering | Silicon Valley, USA | 2022

Battery Labs | Li-ion battery research | Arizona, USA | 2017

Indus Earth Trust | Distributed solar systems | Karachi, Pakistan | 2018

Fauji Fertilizer | Urea process & operations | Mirpur Mathelo, Pakistan | 2014-15

Fatima Fertilizer | DAP process & operations | Sadiqabad, Pakistan | 2014

LAST 5 YEARS



Nanomaterials Lab @ UConn



Racquetball tournament in CT



R&D facility @ Tesla



Fishing in the Atlantic ocean



Black Sand beach, Iceland



Washington, D.C.

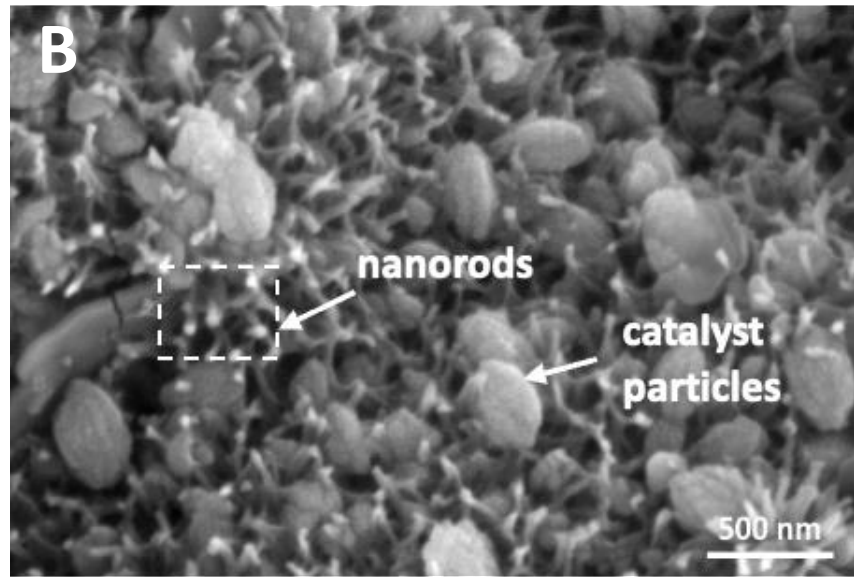
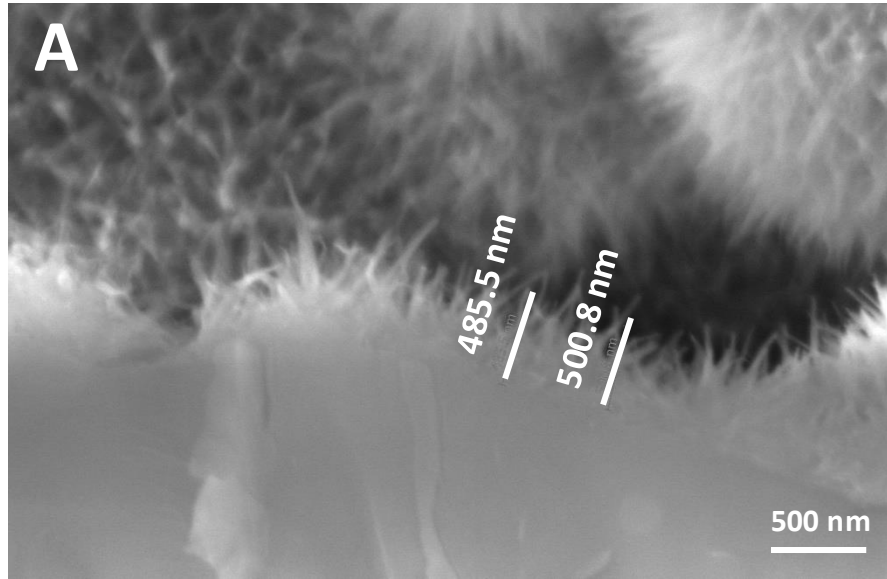


Venice, Italy

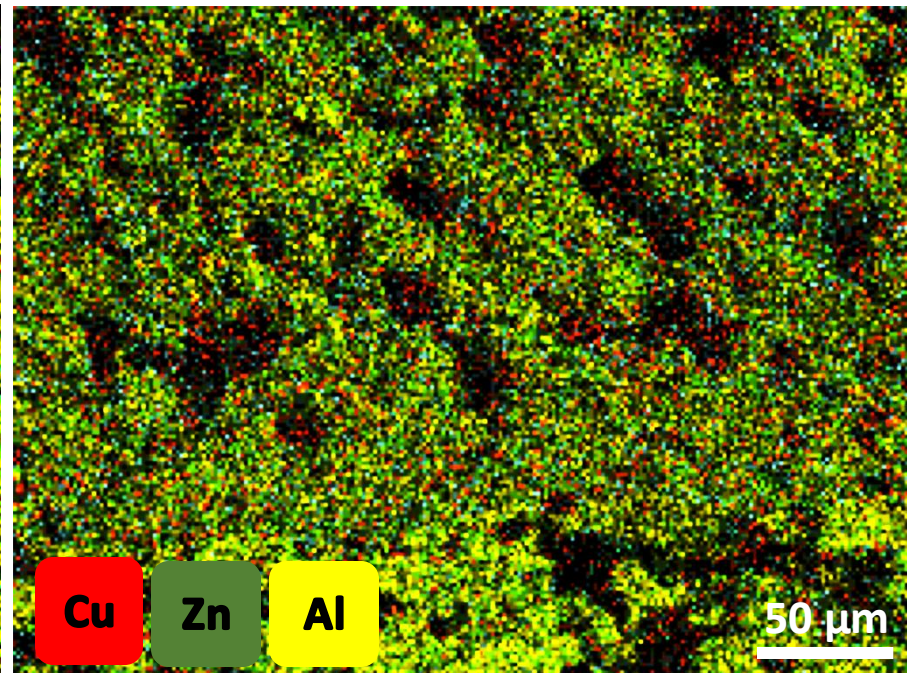
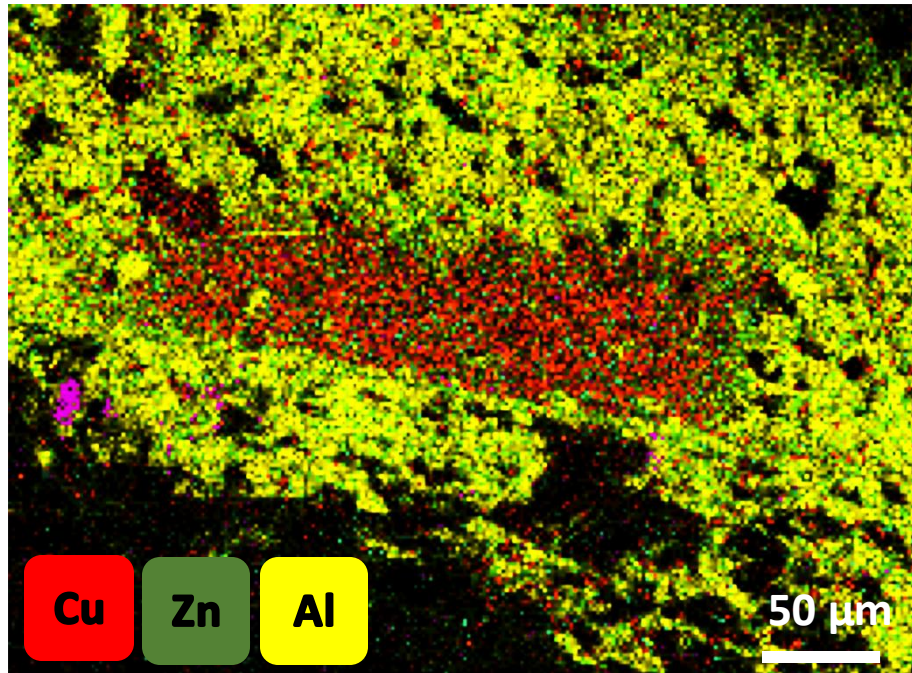
Nanomaterials Lab



Habib University
shaping futures



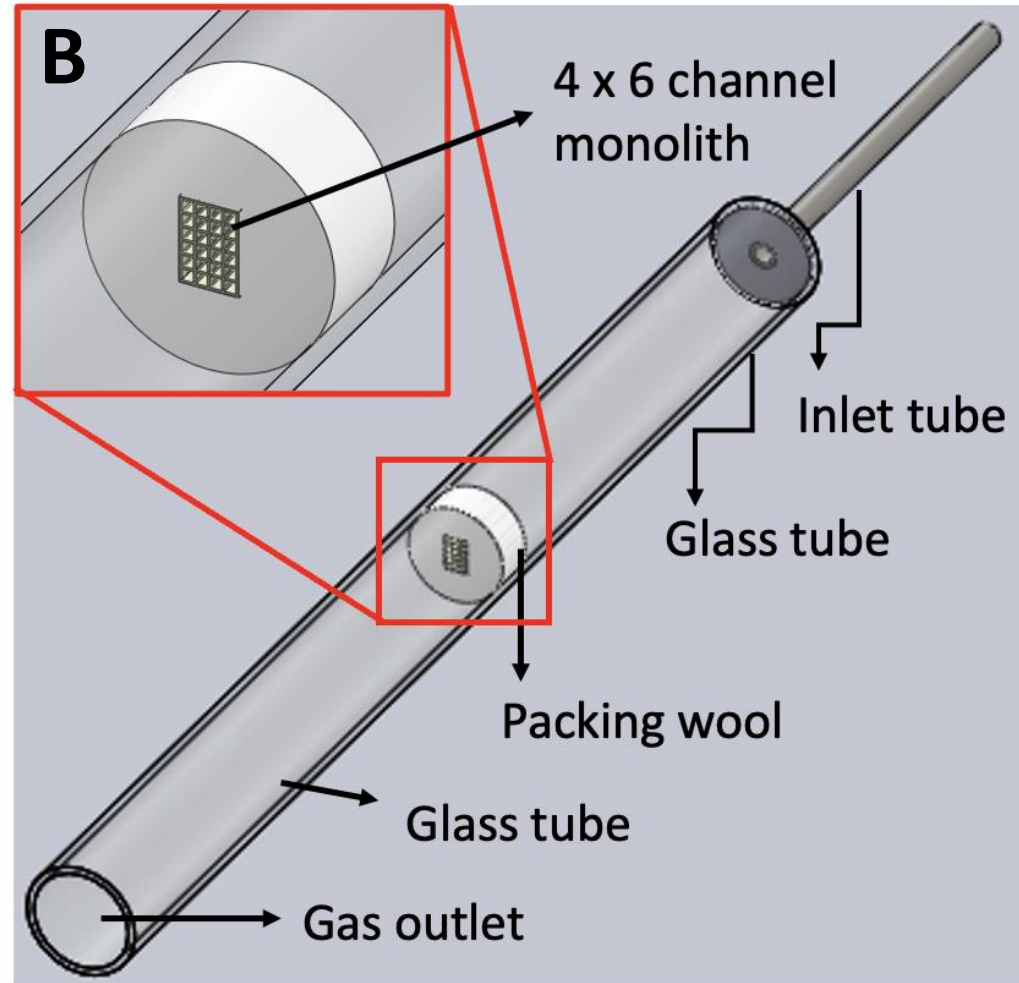
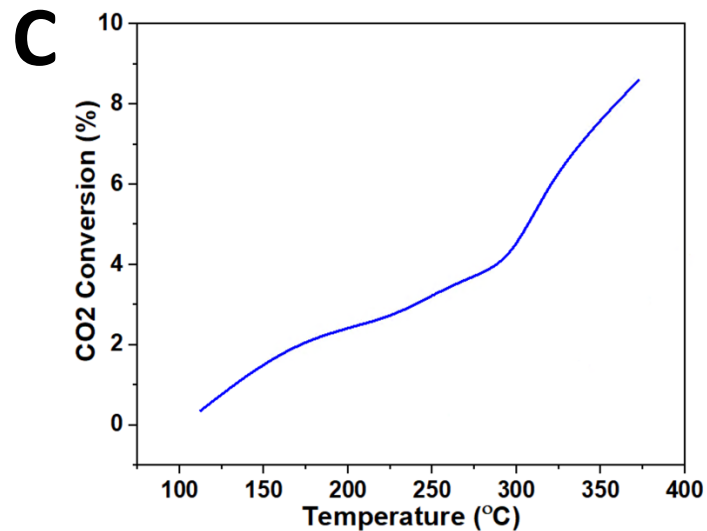
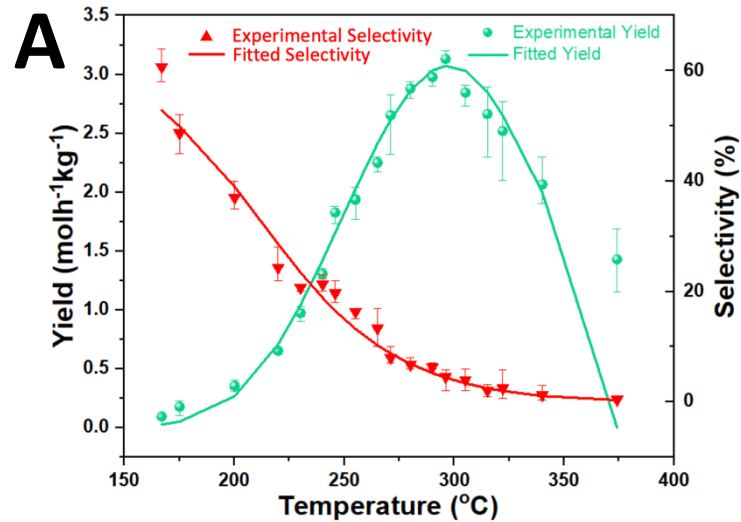
Growing nanomaterials
For catalysis



Nanomaterials Lab



Habib University
shaping futures



Experimental Data
modeling
&
Reactor Simulation

Nanomaterials Lab

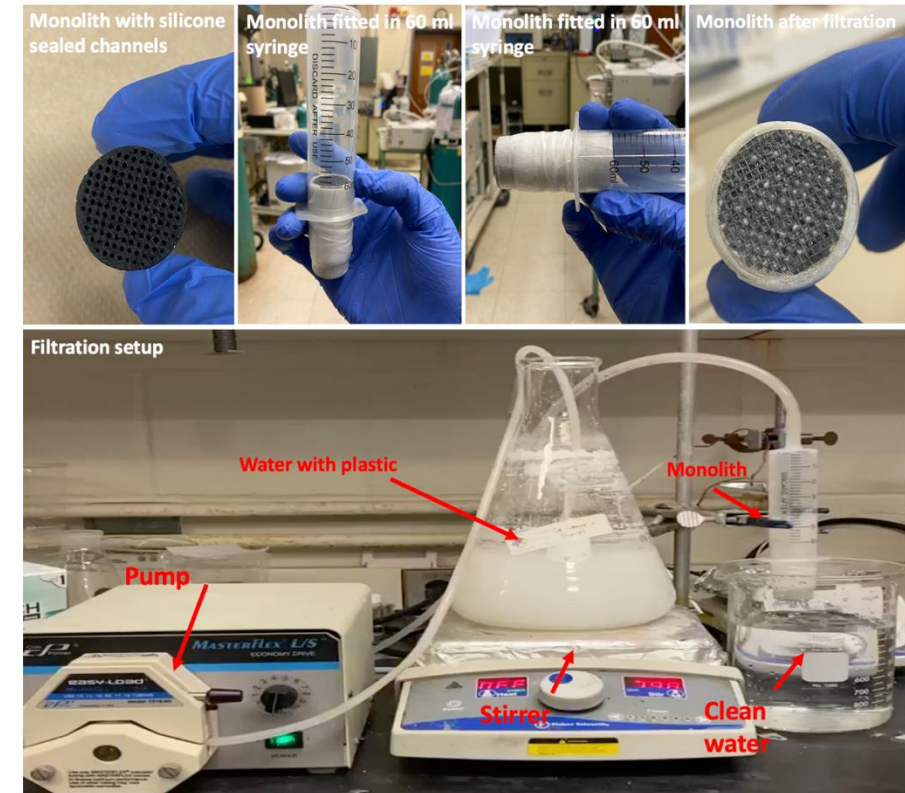
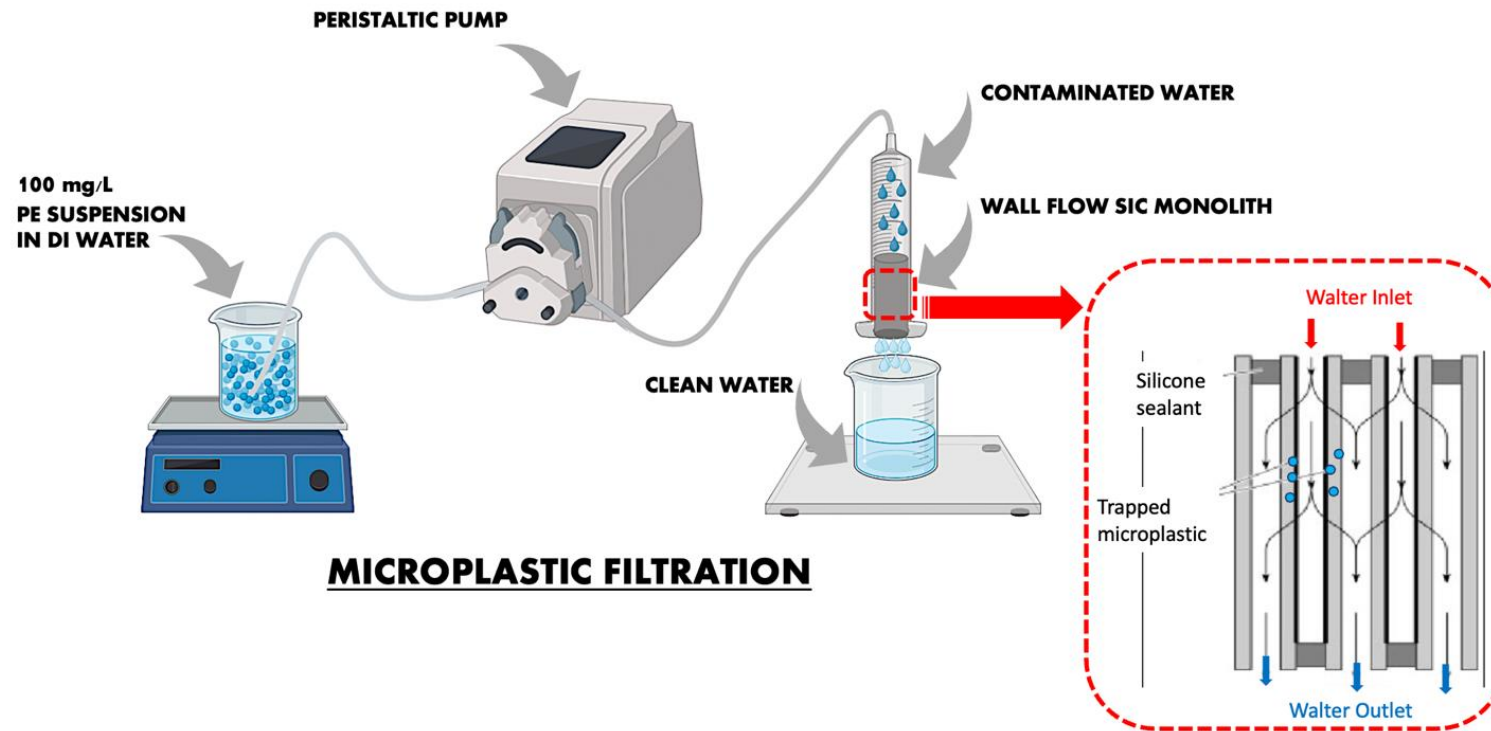
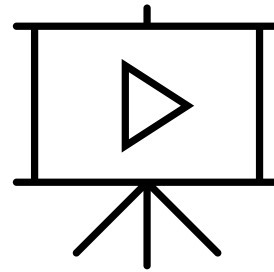


Figure. Microplastic filtration setup schematics [red dotted box depicts wall flow through the nanostructure integrated monolithic filter], drawn using biorender.com

Nanomaterials Lab



Video: microplastic water suspension
filtration using wall-flow SiC monolith

Distributed solar plants

- Small scale solar systems installation and maintenance for remote villages in Thatta, Sindh
- Community awareness and trainings for renewable energy



Source: Indus Earth trust report 2018

Urea Production (FFC)

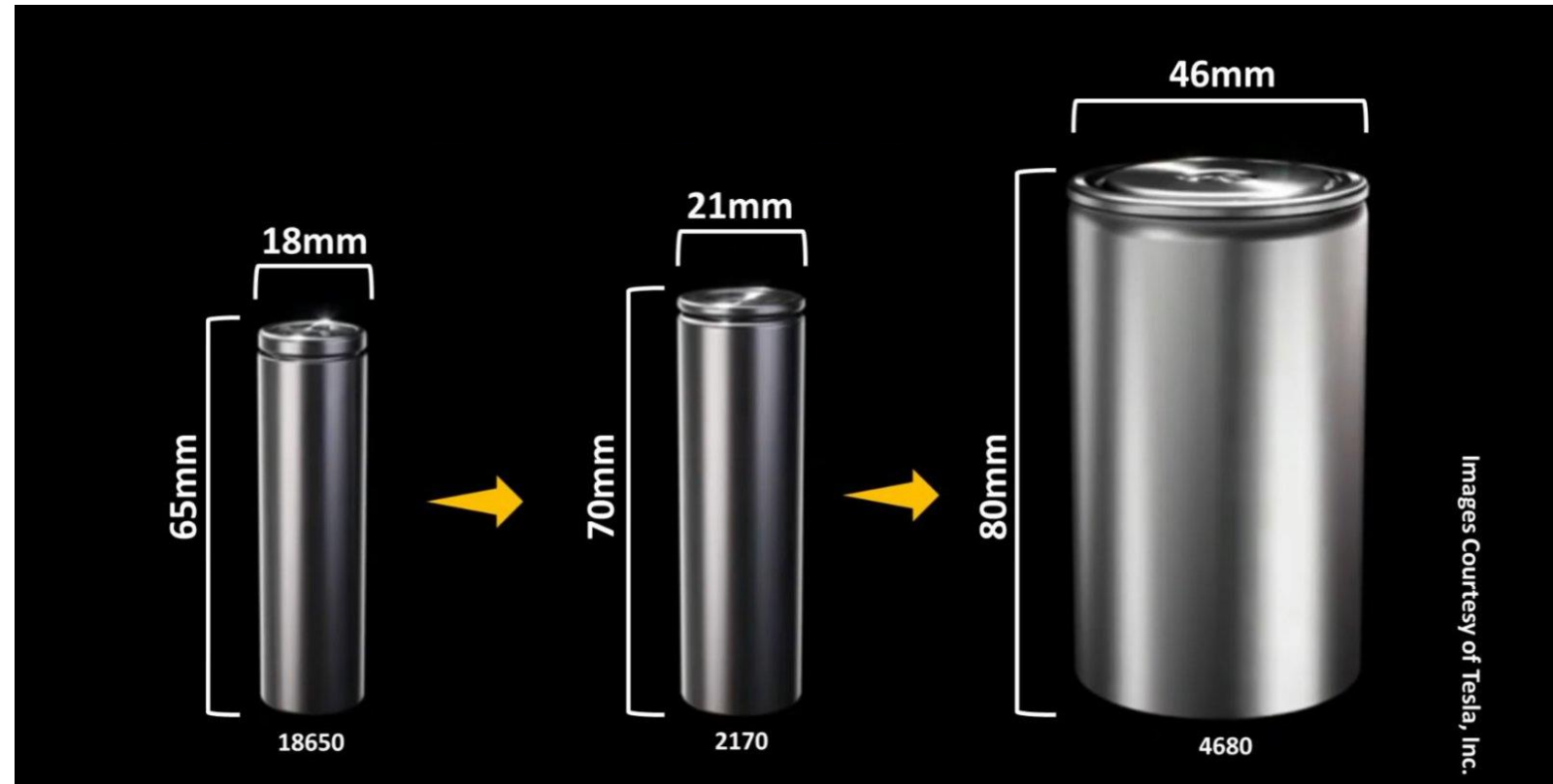
- Sona urea
- Engineering operations and process



Source: FFC Annual report 2021

Tesla Motors

- R&D facility for new 4680 cell
- Solvent free technology



Introduce yourself!

- Name
- Program/Major
- Why did you choose this elective?
- What's the first thing that comes to mind when you think about water? Is it a place, a memory, or a time?



Course Policies

- Canvas is the best place to find course related materials and official announcements.
- 85% attendance is mandatory
- No Plagiarism
- Come prepared to class after completing assigned readings or tasks.
- Listen respectfully to your peers and their diverse viewpoints.
- Maintain a professional and respectful tone in written communication.
- Properly cite sources and give credit when using others' work.
- Ensure everyone has an equal chance to participate by encouraging turn-taking.
- **No N-1**
- **No extra assignments and quizzes**
- **Attendance**

Late Submission Policy

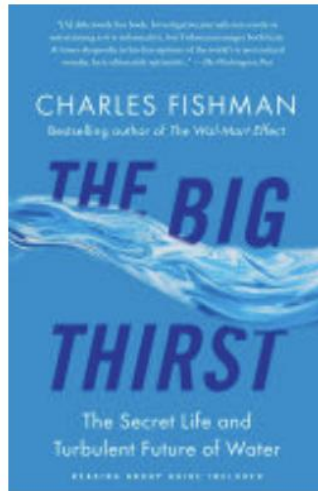
- All assignments will be due before start of class on Canvas. If your assignment is late, you will be docked on the following system:
- 1 day late 20% of the maximum allowable marks
- 2 days late 40% of the maximum allowable marks... and so on



COURSE DESCRIPTION

- "Water: Science, Society, and Policy" delves deeply into water's scientific aspects, societal impacts, and policy frameworks, offering a holistic understanding of its significance.
- Through interdisciplinary study, students will grasp the pivotal role water plays in ecosystems, human societies, and global sustainability.
- The course navigates the conflicts arising from diverse societal water needs and the impact of climate change on hydrology, using global case studies.
- With a focus on Karachi's water system, students will learn stakeholder perspectives and devise equitable management plans, fostering awareness of water challenges and management strategies.

Course Books



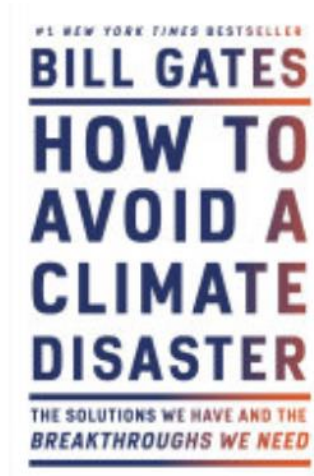
The Big Thirst

ISBN: 9781439124932

Authors: Charles Fishman

Publisher: Simon and Schuster

Publication Date: 2011-04-12



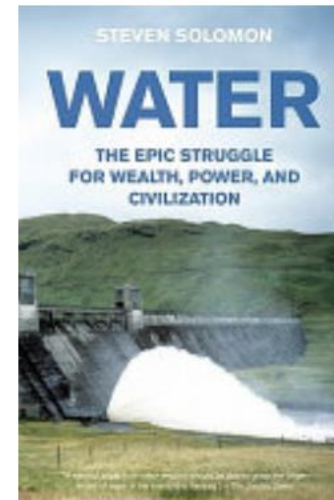
How to Avoid a Climate Disaster

ISBN: 9780735280458

Authors: Bill Gates

Publisher: Knopf Canada

Publication Date: 2021-02-16



Water

ISBN: 9780060548315

Authors: Steven Solomon

Publisher: Harper Collins

Publication Date: 2011-01-18

GRADING

Component	Type	Weightage (%)
Water Journal	Individual	15
Assignments	Individual	10
Quizzes	Individual	(2x10) 20
Discussions groups	Group	12.5
Presentations	Pair	12.5
Research Project	Pair	30



Water Journal (15%)

- Students keep a log of their daily water usage and then use that data to calculate water footprints

Activity	Average Water Usage (liters)
Drinking water	
Showering	
Washing hands	
Cooking	
Flushing toilet	
Brushing teeth	
Laundry	
Other (specify)	
Virtual water (products)	
Total Daily Usage	



Discussion groups (12.5%)

- Class discussion on select topic
- Students will discuss assigned topics in small groups.
- Each group will present their conclusions to the class

Presentations (12.5%)

- Group activity
- Students give presentations on their picked country's water management system

Assignments (10%)

- Assignment will be related to some numerical problems related to water.

Quizzes (20%)

- 2 Quizzes



Research Project (30%)

- Group activity
- Students review literature and conduct field work to understand local water related problems and propose possible solutions.
- Term paper proposal
- Primary data collection
- Secondary research
- Mid term project review, a presentation on the literature findings and project progress.
- Biweekly project updates
- Final report, a comprehensive paper with citation, data and proposed solutions.

Best of luck!